Pro Football Sports Science Research Update

August 2025

A Bayesian circular mixed-effects model for explaining variability in directional movement in American football. arXiv:2507.06122: 1-20, 2025.

An analysis of Guardian Cap use and changes in the concussion rate in National Football League preseason practices from 2018 to 2023. The American Journal of Sports Medicine, Ahead of Print: 1-8, 2025.

Analysis of Z-score and total score of athleticism on drafted and undrafted players from the NFL Scouting Combine. Scientific Reports, 15: 21742, 2025.

Anterior cruciate ligament injuries in National Football League players from 2012 to 2022: A descriptive epidemiology study. The Journal of Knee Surgery, Ahead of Print: 1-25, 2025.

Concussion injuries in sports and the role of instrumented mouthguards: A mini-review. Frontiers in Bioengineering and Biotechnology, 13:1567429, 2025.

Concussions are associated with increases in blood pressure and cardiovascular risk in American-style football athletes. JSCC: Advances, 4(5): 1-12, 2025.

Evidence of ostarine cross-contamination via sweat in 2 athletes sharing the same neoprene hamstring sleeves. An original situation of drug transfer where the anti-doping rule violation was suspended by the sport authorities. Clinica Chimica Acta, 559: 119688, 2024.

Fascia and muscle stiffness in soccer athletes with and without previous hamstring injury. Journal of Functional Morphology and Kinesiology, 10: 48, 2025.

From the Field to the Screen: A scoping review of video analysis as a tool for understanding thigh muscle-tendon injuries in football. Clinical Journal of Sport Medicine, Ahead of Print: 1-12, 2025.

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Full return to sport after suture-button fixation for syndesmotic injuries in all athletes: Faster recovery was seen in NCAA football players compared with National Football League players. Arthroscopy, Sports Medicine, and Rehabilitation, Ahead of Print: 1-21, 2025.

How do they decide? Performance support staff perceptions of decision-making in elite-sport environments. Journal of Strength and Conditioning Research, 39(2): 192-199, 2025.

Impact of cervical and lumbar spine surgeries on National Football League (NFL) player performance and return-to-play outcomes. Cureus, 17(6):e85706, 2025.

Lending a hand in bringing ballcarriers down: An extended study of individual defensive contributions in the National Football League (NFL). SN Computer Science, 6:622, 2025.

NFL Draft Modelling: Loss functional analysis. arXiv:2504.07291v1: 1-14, 2025.

Optimizing athletes' recovery and performance: A review of vibration therapy, compression garments, and massage. Current Sports Medicine Reports, 24(6): 177-185, 2025.

Performance enhancing supplements and cardiovascular health. Current Treatment Options in Cardiovascular Medicine, 27: 29, 2025.

Personalized nutrition for the enhancement of elite athletic performance. Scandinavian Journal of Medicine & Science in Sports, 35:e70044, 2025.

Return to sport assessments after anterior cruciate ligament injury: Which jump landing test is sensitive to an ACL injury history under fatigued or non-fatigued conditions? European Journal of Sport Science, 25:e12317, 2025.

Sensorimotor dysfunction following anterior cruciate ligament injury (Part 1): What is it? How can clinicians assess it? Journal of Orthopaedic & Sports Physical Therapy, Ahead of Print: 1-38, 2025.

Sensorimotor dysfunction following anterior cruciate ligament injury (Part 2): How can clinicians rehabilitate it? 55(7): 1-9, 2025.

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Solving heat stress in sport: Intermittent cold air exposure mitigates deterioration of the autonomic nervous system in American football athletes. Medicine and Science in Sports & Exercise, Ahead of Print: 1-29, 2025.

Statistical analysis and machine learning in college football: Insights from offensive efficiency, yards per play, and clustering of play styles. Generative AI in Creative Industries, Studies in Computational Intelligence, 1208, 2025.

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The 4Rs framework of sports nutrition: An update with recommendations to evaluate allostatic load in athletes. Life, 15:867, 2025.

The association between age of first exposure to American football at a young age and later-life health issues in healthy, community-dwelling adults. Sports Medicine, Ahead of Print: 1-13, 2025.

The impact of localized muscle mass on sprint mechanics during the swing and stance phases of Division 1 collegiate American football players. Journal of Strength and Conditioning Research, Ahead of Print: 1-5, 2025.

The influence of limb dominance on performance-based outcomes after ACL reconstruction. Sport Health, Ahead of Print: 1-8, 2025.

The relationship between relative rest and team performance across competitive seasons in the National Football League. Journal of Strength and Conditioning Research, Ahead of Print: 1-5, 2025.

The role of high-intensity interval training (HIIT) in neuromuscular adaptations: Implications for strength and power development – A review. Life, 15: 657, 2025.

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Training impulse and session ratings of perceived exertion as perceived by athletes and observed by coaches during preseason football camp. Journal of Strength and Conditioning Research, Ahead of Print: 1-9, 2025.

A Bayesian circular mixed-effects model for explaining variability in directional movement in American football. arXiv:2507.06122: 1-20, 2025.

Abstract - Change of direction is a key element of player movement in American football, yet there remains a lack of objective approaches for in-game performance evaluation of this athletic trait. Using tracking data, we propose a Bayesian mixed-effects model with heterogeneous variances for assessing a player's ability to make variable directional adjustments while moving on the field. We model the turn angle (i.e., angle between successive displacement vectors) for NFL ball carriers on both passing and rushing plays, focusing on receivers after the catch and running backs after the handoff. In particular, we consider a von Mises distribution for the frame-level turn angle and explicitly model both the mean and concentration parameters with relevant spatiotemporal and contextual covariates. Of primary interest, we include player random effects that allow the turn angle concentration to vary by ball carrier nested within position groups. This offers practical insight into player evaluation, as it reveals the shiftiest ball carriers with great variability in turning behavior. We illustrate our approach with results from the first nine weeks of the 2022 NFL regular season and explore player-specific and positional differences in turn angle variability.

An analysis of Guardian Cap use and changes in the concussion rate in National Football League preseason practices from 2018 to 2023. The American Journal of Sports Medicine, Ahead of Print: 1-8, 2025.

Purpose: To study the effect of Guardian Cap NXT use on the incidence of concussions in National Football League (NFL) preseason practices from 2018 to 2023.

Study Design: Cohort study; Level of evidence, 3.

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Methods: Using data from the NFL Injury Surveillance System, concussion rates in preseason practices during the 2018 to 2023 preseasons (excluding the 2020 preseason) were compared before and after the NFL required players in certain positions (offensive linemen, defensive linemen, tight ends, and linebackers) to wear the Guardian Cap NXT during practices in 2022. Several secondary analyses, including 2 Poisson regression models, were conducted to evaluate the effects of preseason, player position, practice intensity (pads vs no pads), collision partner, helmet shell involvement, and Guardian Cap NXT requirement.

Results: A total of 139 concussions occurred during the preseason Guardian Cap NXT requirement period in 2018, 2019, 2021, 2022, and 2023. After the Guardian Cap NXT requirement was implemented, the concussion rate in the group of affected positions decreased 54% (P = .001) in the univariate analysis and 62% (P = .004) in the regression model. For concussions involving the helmet shell (as opposed to the facemask) (n = 76), there was no significant association between Guardian Cap NXT use and the concussion rate (P = .122), indicating that the reduction in concussions could not be attributed solely to the energy-absorbing effects of the Guardian Cap NXT.

Conclusion: Required use of the Guardian Cap NXT was associated with a 54% to 62% reduction in the overall incidence of NFL practice concussions. However, when only concussions involving a helmet shell impact were considered, there was no longer a significant relationship between Guardian Cap NXT use and the incidence of concussions.

Analysis of Z-score and total score of athleticism on drafted and undrafted players from the NFL Scouting Combine. Scientific Reports, 15: 21742, 2025.

Abstract - This study aims to evaluate the effectiveness of z-score and Total Score Athleticism (TSA) in distinguishing between drafted and undrafted NFL players, with a focus on different positional groups. We analyzed NFL Scouting Combine data (2000-2024) using z-score and TSA calculated for both the entire sample and specific positional groups (Skill player, big skill player and linemen), with a final sample of 3,446 players included after filtering. Examine the differences between drafted and undrafted players, as well as the differences among the three positional groups. The classification accuracy was evaluated by ROC analysis and binary logistic regression. Significant differences were observed in z-

scores (P < 0.001) and TSA (P < 0.001) between drafted and undrafted players under the two calculation methods across all positions, with drafted players generally outperforming undrafted ones. ROC analysis indicated that all z-scores on NFL SC demonstrate good performance (AUC > 0.6, P < 0.001). Z-score for the 40-yard dash provided the highest AUC (0.637-0.719), while the bench press lowest (0.556-0.631). Position-specific z-score and TSA calculations showed better discriminatory performance compared to the calculation based on the overall sample. This study confirms that both z-scores and TSA are effective in differentiating drafted from undrafted players, with position-specific calculations offering enhanced discriminatory power. The results underscore the importance of tailoring athletic assessments to specific positional needs to improve talent identification. Coaches can understand how z-scores and TSA can be used on National Football League Scouting Combine, and these metrics perform across various positions.

Anterior cruciate ligament injuries in National Football League players from 2012 to 2022: A descriptive epidemiology study. The Journal of Knee Surgery, Ahead of Print: 1-25, 2025.

Abstract - There is a high incidence of anterior cruciate ligament (ACL) tears in National Football League (NFL) players. There is a relative paucity of long-term data characterizing the effects of artificial versus natural playing surfaces on ACL tear incidence. A comprehensive search of publicly available data identified any NFL player who sustained an ACL injury from 2012 to 2022. The publicly available databases allowed for characterization of playing surface, player position, timing of injury in season, mechanism of injury, and player age. Incidence of ACL injuries by playing surface, player position, and position group relative to all other positions or position groups were compared by chi squared test. Correlates of in-game ACL injury were determined by stepwise multivariate logistic regression. There were 520 NFL players who experienced an ACL tear between seasons 2012-13 and 2022-23. There was no significant difference in incidence of ACL tears on artificial turf versus natural playing surfaces (265 versus 255, p=0.670). Wide receivers (p=0.027) were associated with a significantly higher incidence of ACL tears compared to other positions. Receivers (wide receivers and tight ends) (p=0.007) were associated with significantly higher incidence of ACL tears compared to other position groups, while specialists (punters, kickers, and long snappers) (p<0.001) were associated with significantly lower

incidence of ACL tears compared to other position groups. No significant difference between mechanism of injury between playing surfaces was found (p=0.765). More ACL tears occurred before week 9 than on or after week 9 in-season, and the proportion of ACL tears occurring during a game (67%) was significantly higher than those occurring during practice. There was no statistically significant difference in incidence of ACL tears in NFL players from the 2012-13 to 2022-23 seasons based on playing surface (artificial versus natural). Wide receivers had a significantly higher incidence of ACL tears compared to other positions and tears were more common during games and before week 9 in-season.

Concussion injuries in sports and the role of instrumented mouthguards: A mini-review. Frontiers in Bioengineering and Biotechnology, 13:1567429, 2025.

Abstract - Contact sports such as American football, rugby, soccer, and ice hockey involve high-speed, high-impact interactions that frequently result in head acceleration events (HAEs), which can lead to concussions and other forms of traumatic brain injury. HAEs can lead to acute symptoms like dizziness and memory difficulties, as well as more severe, chronic conditions like cognitive decline and chronic traumatic encephalopathy. This mini-review focuses on concussion-related injuries in contact sports, examining their prevalence, impact, and the role of innovative prevention strategies. Particular attention is given to the development of instrumented mouthguards (iMGs), which incorporate real-time sensors to measure and analyze head impacts. Ultimately, this review aims to provide an overview of the role of iMGs on concussion prevention and its evolving landscape, with a focus on the potential of iMG technology.

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Concussions are associated with increases in blood pressure and cardiovascular risk in American-style football athletes. JSCC: Advances, 4(5): 1-12, 2025.

Background: Concussions sustained during American-style football (ASF) participation are common. Whether concussions are associated with cardiovascular risk is unknown.

Objectives: The authors sought to determine whether concussions sustained during collegiate ASF participation lead to increases in systolic blood pressure (SBP) and acquired maladaptive cardiovascular phenotypes.

Methods: In a longitudinal and case-controlled analysis, ASF athletes at 2 National Collegiate Athletic Association Division-I programs were followed for up to 2 years with cardiovascular assessments including BP, echocardiography, and vascular applanation tonometry. ASF athletes who sustained clinically diagnosed concussions were compared to weight and player position-matched controls without concussions.

Results: Thirty-nine ASF athletes who sustained concussions (preconcussion baseline: [SBP] 129.6 ± 13.6 mm Hg) and 39 weight and player position-matched controls (preconcussion baseline: [SBP] 131.6 ± 13.1 mm Hg) were followed for 1 season postconcussion; 14 of these pairs were followed through 2 seasons. After 1 season, increased pulse wave velocity (PWV) (case Δ = 0.4 [0.2] m/s, P = 0.036; control Δ = -0.1 [0.1] m/s, P = 0.38) was observed among cases. Among the case and control pairs followed over 2 seasons, increased SBP (case Δ = 10.1 [3.6] mm Hg, P < 0.01; control Δ = 0.1 [3.2] mm Hg, P = 0.97) and diastolic blood pressure (case Δ = 8.2 [2.9] mm Hg, P < 0.01; control Δ = -4.1 [4.0] mm Hg, P = 0.30) and decreased É (case Δ = -2.8 [0.8] cm/s, P = 0.001; control Δ = -1.0 [0.8] cm/s, P = 0.21) were observed among cases. Over 1 season, concussions were associated with higher PWV (β = 0.33 [95% CI: 0.09-0.56], P = 0.007). Among athletes followed over 2 seasons, concussions were associated with higher PWV (β = 0.42 [95% CI: 0.05-0.78], P = 0.03), diastolic blood pressure (β = 5.89 [95% CI: 1.23-10.54], P = 0.01), LV mass index (β = 11.01 [95% CI: 6.13-15.90], P < 0.001), and lower É (β = -2.11 [95% CI: -3.27 to -0.95], P < 0.001).

Conclusions: Concussions sustained during collegiate ASF participation are independently associated with markers of cardiovascular risk and acquired maladaptive cardiovascular phenotypes. Clinical ASF

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concussion management strategies inclusive of careful BP surveillance may lead to early identification of hypertension.

Evidence of ostarine cross-contamination via sweat in 2 athletes sharing the same neoprene hamstring sleeves. An original situation of drug transfer where the anti-doping rule violation was suspended by the sport authorities. Clinica Chimica Acta, 559: 119688, 2024.

Abstract - The presence of ostarine, a selective androgen receptor modulator (SARM) in an athlete's sample constitutes one of the most frequent anti-doping rules violation. It is possible to challenge this violation but it is the athlete who has to demonstrate he / she is innocent. The conditions to evidence no fault or negligence are mostly based on 2 points: 1. the athlete or his/her legal representative must present verified circumstances of contamination and the source of contamination must be identified; and 2. there must be verified claims by the athlete about the fact that he / she did not knowingly take the prohibited substance, i.e. that the violation was not intentional. During a 2-weeks period, a male athlete tested two times positive for ostarine in urine (<0.1 ng/ml) and he challenged these results. His hair and nail tests returned negative (LOQ at 0.5 pg/mg). He admitted using two neoprene hamstring sleeves of another athlete who confessed abusing ostarine. This was confirmed in his hair (190 pg/mg), his fingernail clippings (780 pg/mg) and his toenail clippings (45 pg/mg). To document the presence of ostarine in the hamstring sleeves and therefore possible drug transfer, the hamstring sleeves were analysed. Ostarine was identified in 12 different selected pieces (about 1 g) of the sleeves at concentrations ranging from 3 to 142 pg/g. Sport authorities (USADA) agreed that the most likely source of contamination was the hamstring sleeves, thus confirming the scenario of drug transfer and gave the athlete a no fault.

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Fascia and muscle stiffness in soccer athletes with and without previous hamstring injury. Journal of Functional Morphology and Kinesiology, 10: 48, 2025.

Background/Objectives: Despite extensive efforts to reduce injuries to the hamstrings, the injury rate among athletes is increasing. The purpose of this study was to examine fascia and muscle stiffness differences between ten soccer players with a previous biceps femoris long head (BF) injury and thirteen controls.

Methods: The shear-wave elastic (SWE) modulus and surface electromyography signal from the semitendinosus (ST) and BF were measured during passive and active knee flexion efforts from 0°, 45°, and 90° knee flexion angles. Anatomical cross-sectional area (CSA) and maximum isometric strength were also obtained.

Results: Analysis of variance showed that the injured group showed significantly greater active ($p < 0.05$) but similar passive SWE modulus of BF and ST fascia and muscle than the uninjured group. Compared to the non-injured group, injured athletes had lower isometric strength and BF anatomical CSA ($p < 0.05$) but similar electromyographic activation amplitude ($p > 0.05$).

Conclusions: The greater fascia stiffness during active submaximal contractions, in comparison to controls, might have an impact on hamstring function in soccer players with BF injuries who returned to play. Injured players may benefit from therapeutic interventions that aim to restore fascia and muscle tissue stiffness.

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From the Field to the Screen: A scoping review of video analysis as a tool for understanding thigh muscle-tendon injuries in football. Clinical Journal of Sport Medicine, Ahead of Print: 1-12, 2025.

Objective: Muscle-tendon injuries are critical setbacks in professional football, significantly affecting player availability and team performance. Understanding these injury mechanisms through video analysis is crucial for developing effective prevention and rehabilitation strategies that enhance player welfare and optimize performance. This review aimed to synthesize data from articles that used video analysis to explore mechanisms of thigh muscle-tendon injuries in football.

Data source: A comprehensive literature search was conducted from 2010 to 2025 using a scoping review methodology. The quality of the included studies was assessed using the Quality Appraisal for Sports Injury Video Analysis Studies (QA-SIVAS) scale.

Main results: The review of 10 studies identified that noncontact and indirect contact mechanisms are predominantly responsible for severe thigh muscle-tendon injuries in football. Common injury scenarios involved sprinting-induced strains and kicking actions, highlighting the significant role of eccentric loading and rapid biomechanical changes.

Conclusions: Video analysis has emerged as a vital tool in sports medicine, providing deep insights into the complex mechanisms of thigh muscle-tendon injuries in football. The continuous improvement of analytical methods, including the adoption of advanced technologies such as artificial intelligence, is imperative for refining prevention and rehabilitation protocols.

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Full return to sport after suture-button fixation for syndesmotic injuries in all athletes: Faster recovery was seen in NCAA football players compared with National Football League players. Arthroscopy, Sports Medicine, and Rehabilitation, Ahead of Print: 1-21, 2025.

Purpose: The purpose of this study was to determine the rate and time to return to sport (RTS) for NCAA Football and NFL players following suture-button fixation for the treatment of unstable tibiofibular syndesmotic injuries and to identify possible risk factors associated with a delay in RTS.

Methods: NCAA and NFL players who underwent suture-button fixation for an isolated syndesmotic ankle injury between 2017 and 2023 were evaluated. All player information and data were retrieved from publicly available internet sources. Information was gathered on each player's career, NFL participation, number of seasons played before and after surgery, number of games played before and after surgery, and time between surgery and RTS. RTS is defined as time until first game after syndesmotic suture-button fixation.

Results: Seven NCAA and fourteen NFL players (21 ankles) were included. The mean age at surgery was 24.90 ± 4.66 years (range, 18-34 years). A total of 100% of patients were able to RTS at their respective level of sport. The mean time between surgery and the first game played was 139.4 ± 123.6 days (range, 12 - 427 days). Players played in 23.48 ± 21.9 games after surgery (range, 3-72). NCAA players were able to RTS significantly earlier than NFL players (39.0 ± 20.4 vs 189.6 ± 123.2 days, respectively, P = 0.005, with a p-value of 0.05 denoting significance). There was no significant difference in workload following surgery for both NCAA and NFL players.

Conclusion: In this study, we found that athletes, regardless of whether they were NCAA or NFL players, were able to successfully return to their respective sports after ankle surgery. While NCAA players returned to play significantly faster, both groups were able to achieve similar performance levels postoperatively in terms of the number of games played.

Level of evidence: Level III, retrospective cohort study

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How do they decide? Performance support staff perceptions of decision-making in elite-sport environments. Journal of Strength and Conditioning Research, 39(2): 192-199, 2025.

Abstract - Decision-making is a critical component of performance support staff responsibilities, yet research exploring staff perceptions relating to key facets of decision-making is lacking. This study aimed to explore perceptions of performance support staff relating to; the importance of decision-making in their role, the value of team decision-making, strategies to enhance decision-making skills, monitoring the process and outcome of decisions, and the obstacles to good decision-making within their professional contexts. A mixed-methods approached was used to explore perceptions of decision-making in 191 performance support staff (male staff = 179, female staff = 12) working in professional (n = 147) or international (n = 44) sport. Performance support staff reported decision-making as extremely important to their role; however, only 31% had received formal decision-making education and revealed underdeveloped decision-making capabilities within their organizations as a barrier to good decision-making. Most performance support staff conveyed that team decision-making is a valuable endeavor and augments decision outcomes; yet articulated dysfunctional team dynamics as an obstacle to good decision-making in their environments. Largely, performance support staff reported an absence of organizational strategy and workflow systems and processes for tracking and making decisions, respectively, whereas only 22% had a consistent reviewing and updating process. When making decisions, performance support staff faced several different cognitive biases and were challenged by the environmental pressures. The perspectives documented here suggest that performance support staff would likely benefit from undertaking evidence-based training and implementing scientifically informed strategies to enhance decision-making processes within their organizations.

Practical Applications - In professional sports environments similar to those referenced by the subjects in the present study, findings suggest that performance support staff may benefit from education and training to enhance decision-making processes in their organizations. First, to create interest in decision-making education and enhance decision-making skills of practitioners, education on basic relatable psychological concepts may be an appropriate first step. More advance training may then be provided to overcome cognitive biases through evidence-based interventions. For example, Chang et al. (8) demonstrated that a single 45-minute training session in probabilistic-style thinking consistently improved forecasting accuracy. Second, to develop collaboration and overcome identified obstacles to

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team decision-making, evidence-based strategies, such as building psychological safety to overcome poor communication (10,34), may be useful. In addition, the regular inclusion of staff members outside of the normal performance support staff interdisciplinary team may mitigate against conformity and group think (13). Finally, to improve the monitoring of the process and outcome of decisions, strategies such as appointing devil's advocates as critical evaluators (43), and using checklists to ensure comprehensive information review (39), may enhance thoughtful updating of decisions. Although providing novel insights, this study has some limitations. Non-probabilistic sampling (purposive and snowball sampling) was used to recruit subjects. Subsequently, findings may not generalize to all professional sport contexts. Second, although the authors attempted to recruit female subjects, only 12 of the 179 subjects were female. Thus, findings are gender biased. Future research in this domain may wish to adopt a more targeted strategy to engage female subjects. This is the first study to investigate performance support staff's perceptions relating to key facets of decision-making in professional sport. Although, performance support staff recognized the importance of decision-making in their roles and the value that making decisions in teams can bring, they also described the presence of underdeveloped decision-making capabilities within their organizations and significant obstacles to working within teams. Furthermore, performance support staff conveyed an absence of formal strategies and processes for decision-making at a personal and organizational level. When making decisions, performance support staff also faced several different cognitive biases, a lack of decision-making skills in others, and were challenged by environmental pressures. The evidence suggests that performance support staff may benefit from undertaking evidence-based training to improve decision-making skills. Moreover, implementing scientifically informed strategies to overcome obstacles may enhance decision-making processes at a personal, team, and organizational level.

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Impact of cervical and lumbar spine surgeries on National Football League (NFL) player performance and return-to-play outcomes. Cureus, 17(6):e85706, 2025.

Introduction: American football players face a higher risk of spine injuries due to the sport's high-impact nature, especially in the lumbar and cervical spine regions. These injuries may require surgical interventions aimed at allowing athletes to return to the sport. However, the effects of these surgeries on players' performance and career longevity have yet to be comprehensively studied.

Objective: This study aims to evaluate the impact of spine surgeries on National Football League (NFL) players' return-to-play rates and performance. We hypothesize that players undergoing lumbar surgeries would demonstrate greater performance improvement and return-to-play rates compared to those undergoing cervical surgeries, with differences influenced by player position and injury location. Study design and methods: This is a retrospective cohort study (III) for which NFL injury reports from 2005 to 2022 were reviewed to identify players who had undergone spine surgery. Data collected included player position, return-to-play, and years played following the procedure. Performance metrics were gathered using Super Bowl wins and Pro Football Focus (PFF) player performance ratings. Statistical analysis was conducted using Python version 3.10.12 (Python Software Foundation, Wilmington, DE, USA) to evaluate differences in return-to-play rates, performance changes, and career duration post-surgery.

Results: The study identified 144 spine surgeries (77 lumbar, 67 cervical) among 136 players. Players who had lumbar surgery had a 61% return-to-play rate, with an average performance rating increase of 6.3%. In contrast, those who had cervical surgery had a 47% return-to-play rate and an average performance rating decrease of 5.8%. Lumbar surgeries were more common among linemen with higher BMIs, while cervical surgeries were more frequent in skill positions. Players with a history of lumbar surgeries were more likely to return to play than those without previous surgeries.

Conclusion: Spinal surgeries significantly impact the careers of NFL players. Lumbar surgeries show better outcomes in terms of return-to-play rates and performance improvements compared to cervical surgeries. The differences in surgical outcomes based on injury location and player position highlight the need for tailored rehabilitation protocols. This study provides valuable insights for medical practitioners, team management, and athletes, contributing to a broader understanding of the implications of spine surgeries in professional football.

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Lending a hand in bringing ballcarriers down: An extended study of individual defensive contributions in the National Football League (NFL). SN Computer Science, 6:622, 2025.

Abstract - In this paper we examine game data from the National Football League (NFL) to uncover insights on how defensive teammates collectively bring down offensive ballcarriers. Traditional defensive statistics focus on individual tacklers, which is certainly vital, but in a team game more than one defender often contributes to a stop. We found that individuals who slowed down ballcarriers had a strong positive correlation with tackles. We also found that the distance that defensive players were from ballcarriers was a statistically significant predictor of ballcarrier speed. We were particularly interested in the influence that defender distances had on ballcarrier speed deceleration, leading to a tackle. We conclude with recommendations for coaches, agents, and front office executives. This paper aims to contribute to the sports analytics topics of player evaluation and performance analysis using player-tracking data, regression modeling, and data science techniques.

NFL Draft Modelling: Loss functional analysis. arXiv:2504.07291v1: 1-14, 2025.

Abstract - In the NFL draft, teams must strategically balance immediate player impact against long-term value, presenting a complex optimization challenge for draft capital management. This paper introduces a framework for evaluating the fairness and efficiency of draft pick trades using norm-based loss functions. Draft pick valuations are modelled by the Weibull distribution. Utilizing these valuation techniques, the research identifies key trade-offs between aggressive, immediate-impact strategies and conservative, risk-averse approaches. Ultimately, this framework serves as a valuable analytical tool for assessing NFL draft trade fairness and value distribution, aiding team decision-makers and enriching insights within the sports analytics community.

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Optimizing athletes' recovery and performance: A review of vibration therapy, compression garments, and massage. Current Sports Medicine Reports, 24(6): 177-185, 2025.

Abstract - It is evident that vibration therapy, massage, and compression garments are increasingly common recovery modalities among amateur and elite athletes. We performed a literature review via PubMed examining the effect of these modalities on measures of sports performance and recovery. We identified 29 eligible articles for vibration therapy, 31 eligible articles for compression garments, and 25 eligible articles for massage. Our review revealed varying degrees of support for each of the recovery modalities. However, our review also highlighted a need for standardized protocols in future studies to better understand how the modalities help athletes recover as to define optimal duration and frequency for maximal effects. Overall, all modalities reviewed seem safe to use and likely would not hinder sports performance or prolong recovery. It is a matter of how much benefit an athlete can gain when recovering from training.

Performance enhancing supplements and cardiovascular health. Current Treatment Options in Cardiovascular Medicine, 27: 29, 2025.

Purpose of Review: This review explores the growing use of performance enhancing supplements (PES) with a focus on those used among recreational athletes. We discuss the mechanisms of action, benefits, and cardiovascular risks associated with these substances to inform safe practices and address an important area in sports cardiology.

Recent Findings: Anabolic–androgenic steroids (AAS), used to supplement testosterone levels beyond physiological levels, are increasingly popular among recreational athletes, with their misuse linked to serious cardiovascular and metabolic disease, including dyslipidemia, hypertension, arrhythmias, coronary artery disease, and cardiomyopathy. Recent large observational data suggest a more than twofold increase in mortality risk among AAS users. By contrast, randomized clinical data suggest relative cardiovascular safety of testosterone replacement therapy (TRT) in older hypogonadal men, despite a mildly elevated risk of thromboembolic events and arrhythmias. For postmenopausal women,

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hormonal therapy (HT) may offer improvements in exercise performance and possibly cardiovascular health when initiated early but may increase cardiovascular and thrombotic risk in older women or those with comorbidities. Beyond AAS and hormonal therapies, pre-workout supplements and creatine are widely used among athletes to improve focus, strength, and aid in recovery. Although their excessive use raises concerns, low to moderate consumption of caffeine and creatine appears to carry minimal cardiovascular risk.

Summary: PES are increasingly prevalent in sports, offering potential performance benefits but also posing significant cardiovascular risks when misused. Each PES category has unique mechanisms of action and safety profiles. Raising awareness of PES and their cardiovascular risk among athletes and sports cardiologists is critical to mitigate adverse events and maintain athlete health.

Personalized nutrition for the enhancement of elite athletic performance. Scandinavian Journal of Medicine & Science in Sports, 35:e70044, 2025.

Abstract - Enhancing athletic performance through the manipulation of nutritional intake has ancient roots, with early guidance from "philosophical giants" like Hippocrates, who describes the balance between diet and exercise. Modern sports nutrition emerged in the 20th century, with research identifying carbohydrate (CHO) intake as beneficial for endurance. Studies like Gordon's in the 1920s linked blood glucose levels to marathon performance, while Cade's research in the 1960s on fluid and electrolyte intake led to the founding of Gatorade and the shift toward drinking during exercise to allegedly prevent dehydration and improve sporting performance. Today, sports nutrition is in a "holding pattern" after significant developments in the 1980s, 1990s, and the 2000s. A new era will involve personalized nutrition, but this development will require a game-changing injection of momentum, recognizing that athletes' responses to nutrition interventions vary widely. New technologies will also need to be developed and perfected, including wearables for real-time biometric monitoring (e.g., heart rate variability, glucose, and sweat composition and rate), which offer potential for tailored nutrition (i.e., diet and hydration) strategies. Applications of genetic and multi-omics technologies (like genomics, transcriptomics, metabolomics, proteomics, and epigenomics) are needed to unlock the potential of personalized sports nutrition by analyzing individual responses to factors such as sleep, nutrition, and exercise. The future lies in fast integration of all available data using next-

generation bioinformatics and AI to generate personalized recommendations, with an emphasis on empirical evidence rather than solely commercial interests. As technology matures, sports (and exercise) nutrition will continue refining its practices but will need a paradigm shift to deliver precise interventions that may offer athletes the crucial edge needed to maximize performance while promoting short-term and long-term health.

Return to sport assessments after anterior cruciate ligament injury: Which jump landing test is sensitive to an ACL injury history under fatigued or non-fatigued conditions? European Journal of Sport Science, 25:e12317, 2025.

Abstract - Accurately identifying residual biomechanical deficits following an anterior cruciate ligament injury is critical for effective rehabilitation and safe return to sport. This study aimed to determine which of four jump-landing tasks demonstrated the greatest sensitivity in distinguishing individuals with a history of ACL injury from healthy controls. Forty-three participants formed the ACL (n = 21, 11 females) and the control group (n = 22, 12 females). Three-dimensional motion data (Vicon, 250 Hz) were recorded during a single-leg hop, unilateral countermovement jump, unilateral crossover hop, and medial-rotation hop before and after a fatigue-inducing intervention (single-leg squats and step-ups). Logistic regression models to classify participants were built using 13 lower-body, trunk, and pelvis joint angles at 50 ms after initial ground contact, angular changes in these angles between 50 and 80 ms, and principal components derived from these variables. Classification rates and individual classification outcomes were assessed. The results revealed that no single jump-landing task consistently outperformed others in detecting ACL injury history. Classification outcomes were influenced by fatigue state and analytical approaches. Fatigue was found to enhance classification rates. Combining joint angles with their temporal changes improved classification rates compared to using joint angles alone. However, applying principal component analysis as a preprocessing step did not consistently enhance model performance. Overall, the study demonstrated that jump-landing tasks, combined with a variety of analytical approaches, can effectively detect ACL injury history. Fatigue enhanced classification outcomes, suggesting that it amplifies differences between post-injury and healthy movement characteristics.

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Sensorimotor dysfunction following anterior cruciate ligament injury (Part 1): What is it? How can clinicians assess it? Journal of Orthopaedic & Sports Physical Therapy, 55(6): 1-17, 2025.

BACKGROUND: The anterior cruciate ligament (ACL) plays an important sensory role within the sensorimotor system. Following ACL injury, sensorimotor dysfunction can have implications for rehabilitation and risk of reinjury.

CLINICAL QUESTION: What dysfunction occurs within the sensorimotor system following ACL injury, and how can clinicians assess it?

KEY RESULTS: Following ACL injury, dysfunction occurs across the sensorimotor system: afferent pathways, efferent pathways, and central processing. The afferent pathways exhibit dysfunction within the somatosensory system ([1] increased pain and swelling, [2] increased central cortical processing, and [3] reduced proprioception). There is also dysfunction in the visual system (increased visual-motor reliance and central cortical processing). The efferent pathways have reduced excitability of the central cortex, reduced descending motor pathway excitability and altered spinal reflexive excitability (acutely reduced but then chronically increased).

CLINICAL APPLICATION: Protocols to assess athletes' sensorimotor function following ACL injury might help clinicians quantify the risk of reinjury. Assessing central processing requires specialized equipment, not typically accessible to clinicians. A practical approach to quantify the extent of sensorimotor dysfunction could focus on assessing the afferent and efferent pathways: tests of proprioception (eg, joint position sense test), pain (eg, visual analog scale and numerical pain rating scale), swelling (eg, sweep test and ballottement test), visual-motor reliance (eg, stepdown test), visual-motor processing ability (eg, sensory stations or neurocognitive tests), muscle strength (eg, repetition maximum testing or isokinetic dynamometry), and voluntary activation (eg, electromyography).

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Pro Football Sports Science Research Update

Sensorimotor dysfunction following anterior cruciate ligament injury (Part 2): How can clinicians rehabilitate it? 55(7): 1-9, 2025.

BACKGROUND: Sensorimotor dysfunction following anterior cruciate ligament (ACL) injury and subsequent ACL reconstruction (ACLR) may lead to errors in motor planning and execution. Dysfunction occurs even after athletes have completed rehabilitation. Therefore, clinicians need to implement strategies to improve how sensorimotor dysfunction is addressed during rehabilitation.

CLINICAL QUESTION: How can clinicians improve their rehabilitation of sensorimotor dysfunction following ACLR? The aim of this clinical commentary is to review methods that may improve rehabilitation by targeting sensorimotor dysfunction following ACL injury.

KEY RESULTS: Rehabilitation should focus on 2 priorities: (1) improving peripheral and central efferent function, and (2) improving somatosensory function and reducing reliance on the visual-motor system.

CLINICAL APPLICATION: Somatosensory function can improve with proprioceptive training but should be implemented within the first 6 weeks of injury/surgery to minimize the chance of increased reliance on the visual-motor system. Methods to increase the complexity of proprioceptive training includes varying the (1) type of task (eg, jumping, decelerating, etc), (2) the visual information used during the task, (3) the cognitive loading during the task, and (4) perturbations. Visual distraction training may be useful for challenging an athlete during sports-specific training and reduce the reliance on the visual-motor system. Improving peripheral and central efferent function involves using regular cryotherapy and transcutaneous electrical nerve stimulation in the early stages to minimize pain and improve muscle recruitment. Throughout rehabilitation, strength training, neuromuscular electrical stimulation, and surface electromyography biofeedback methods improve central and peripheral efferent function when prescribed at a high intensity.

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Solving heat stress in sport: Intermittent cold air exposure mitigates deterioration of the autonomic nervous system in American football athletes. Medicine and Science in Sports & Exercise, Ahead of Print: 1-29, 2025.

Objectives: Cooling interventions mitigating heat-related declines in autonomic nervous system (ANS) function are understudied in collegiate American football.

Methods: This study exposed 50 collegiate male athletes during their 2023 season to two bouts of cold air via air-conditioning before (60 min) and during (5 min) training sessions in a hot and humid climate. Armband monitors (Warfighter Monitor™; Tiger Tech Solutions, Inc., Miami, FL) were used to continuously measure body temperature (°C) during training and baseline heart rate (HR), HR recovery and HR variability (HRV) 24 h post-training. Two HRV time-domain indices were estimated: root mean square of the standard deviation of the NN interval (rMSSD) and the standard deviation of the NN interval (SDNN). These data (2023) were then compared with data collected throughout the previous season (2022) during which players were not provided a cold-air intervention. As such, the 2022 cohort served as a "control" group.

Results: Linear regression models no statistically significant associations at the following body temperature thresholds: ≥37.0°C (baseline HR: P = 0.74; HR recovery: P = 0.71, rMSSD: P = 0.74; SDNN: P = 0.78), ≥38.0°C (baseline HR: P = 0.74; HR recovery: P = 0.74; rMSSD: P = 0.68; and SDNN: P = 0.89), and ≥39.0°C (baseline HR: P = 0.75; HR recovery: P = 0.82; rMSSD: P = 0.78 and SDNN: P = 0.83). Compared with the 2022 season, the athletes spent considerably less time training under the higher body temperature thresholds (≥38°C: 19.5 vs 9.7 min; ≥39°C: 10.5 vs 0.9 min), respectively, despite longer overall training duration (161.1 vs 187.4 min).

Conclusions: Our findings suggest that incorporating bouts of cold air exposure when training in high heat and humidity may mitigate the established heat-related deterioration in ANS function.

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Pro Football Sports Science Research Update

Statistical analysis and machine learning in college football: Insights from offensive efficiency, yards per play, and clustering of play styles. Generative AI in Creative Industries, Studies in Computational Intelligence, 1208, 2025.

Abstract - In recent years, data analytics has transformed college football by providing teams with insights into performance metrics that influence game outcomes. This study investigates the role of offensive and defensive metrics in predicting team success in college football, focusing on offensive efficiency, yards per play, and turnover margin. The problem addressed is understanding which statistical indicators are most strongly correlated with team wins and how these metrics can be used to classify play styles and predict score margins. Using a dataset covering college football seasons from 2013 to 2023, this analysis employs statistical methods, machine learning, and clustering to evaluate the impact of various performance metrics. The methodology includes descriptive analysis to identify top-performing teams, correlation analysis to uncover relationships between metrics and win rates, Ordinary Least Squares (OLS) regression to model key predictors of team success, K-Means clustering to categorize teams by play style, and Random Forest regression to predict score margins based on offensive and defensive factors. The results reveal that offensive efficiency and yards per play are the most significant predictors of wins, with a noticeable upward trend in these metrics across the decade. High-performing teams, such as Oklahoma, Alabama, and Ohio State, consistently rank at the top in offensive efficiency, suggesting the critical role of offensive power in achieving game success. Clustering analysis identifies three distinct play styles, separating high-offensive-efficiency teams from those with more conservative strategies. The predictive model for score margins, based on Random Forest regression, achieves a root mean square error (RMSE) of 6.82, with offensive metrics contributing the most to score predictions. These findings imply that college football teams can enhance their performance by focusing on offensive efficiency, optimizing yardage per play, and maintaining strong turnover margins. This research provides a framework for data-driven strategy development in college football, allowing teams to benchmark against top performers and adjust their tactics based on empirical insights.

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Tackle height and tackle success – An analysis of 52,204 tackle events. European Journal of Sport Science, 25:e70003, 2025.

Abstract - To compare the probability of tackle success (the tackler preventing the ball-carrier and ball from progressing towards the tackler try-line) when contacting the ball-carrier at different heights (shoulder, mid-torso and legs) for different types of tackles (active, passive, smother and arm) while accounting for other tackler situational factors within seven playing levels. Video footage of 271 male rugby union matches were analyzed across seven playing groups (Under [U] 12, n = 25 matches; U14, n = 35; U16, n = 39; U18 Amateur n = 39; U18 Elite n = 38; Senior Amateur, n = 40 and Senior Elite, n = 50) across England, New Zealand, South Africa, Portugal and USA (a total of 51,106 tackles). A multi-level logistic regression model with tackle success as the outcome variable and first point of contact and type of tackle as the explanatory variables were computed. Included in the model as cofounders were the situational variables tackle direction, tackle sequence, number of players in the tackle and attacker intention. Post-estimation marginal effects were used to calculate the probabilities (expressed as a percentage %) of tackle success for each interaction between tackle type (active shoulder, smother, passive shoulder and arm) and the first point of contact (shoulder, mid-torso and legs). The probability of tackle success in relation to where the ball-carrier is contacted varied by tackle type and within each age group. The probabilities (Pr) for contacting the shoulder versus mid-torso at the senior levels (elite and amateur) did not differ in relation to tackle success (for instance, for active shoulder tackles within senior elite; shoulder Pr 86% 95% CI 82-89 and mid-torso Pr 82% 95% CI 77-86), whereas at the junior levels, contacting the shoulder had a higher probability than other points of contact. Active shoulder tackles had the highest probability of tackle success across the different playing levels across the different contact heights, whereas arm tackles had the lowest probability (for instance, for mid-torso tackles within senior elite, active Pr 82% 95% CI 77-86 vs. arm Pr 69% 95% CI 64-75). Coaches and practitioners can use this information to improve tackle training design and planning within the different age groups and facilitate player development.

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The 4Rs framework of sports nutrition: An update with recommendations to evaluate allostatic load in athletes. Life, 15:867, 2025.

Abstract - The 4Rs of sports nutrition were proposed in recent years as an evidence-based framework to optimize post-exercise recovery within the context of allostasis. Under this paradigm, it is important to consider that each R represents a factor with a tremendous influence on the allostatic response and improves individual components of the allostatic load (AL), which will positively impact the exercise-induced adaptations and the athlete's recovery. The 4Rs correspond to the following. (i) Rehydration-This is necessary to guarantee the post-exercise consumption of at least 150% of the body mass lost during the exercise accompanied by sodium (if faster replacement is required). (ii) Refuel-Carbohydrate intake (~1.2 g/kg body mass per hour for up to 4 h post-exercise) is essential not only in restoring glycogen reserves but also in supporting the energy needs of the immune system and facilitating tissue repair. Despite changes in substrate utilization, a ketogenic diet generally has neutral or negative effects on athletic performance compared to carbohydrate-rich diets. (iii) Repair-The ingestion of high-quality protein stimulates post-exercise net muscle protein anabolism and might contribute to faster tissue growth and repair. The use of certain supplements, such as creatine monohydrate, might help to enhance recovery, while tart cherry, omega-3 fatty acids, and dietary nitrate (e.g., Beta vulgaris, Amaranthus L.), as well as other herbal extracts containing flavonoid-rich polyphenols, deserve further clinical research. (iv) Recuperate-Pre-sleep nutrition (casein- or protein-rich meal with slow digestion rate) has a restorative effect, facilitating the recovery of the musculoskeletal, endocrine, immune, and nervous systems. In this article, we update the 4Rs framework, delve deeper into the allostasis paradigm, and offer theoretical foundations and practical recommendations (the 4Rs app) for the assessment of AL in athletes. We cautiously propose an AL index (ALindex) for physique competitors and elite athletes to evaluate the cumulative physiological stress induced by exercise and, thereby, to adjust exercise and nutrition interventions.

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The association between age of first exposure to American football at a young age and later-life health issues in healthy, community-dwelling adults. Sports Medicine, Ahead of Print: 1-13, 2025.

Introduction: Younger age of first exposure (AFE) to American Football (football) is associated with later-life health problems among former professional athletes in several studies; however, studies examining amateur (i.e., nonprofessional) athletes are less clear.

Objective: In a cohort of former amateur American Football players, this study assessed whether AFE to football was associated with: (1) psychiatric and neurobehavioral symptoms, (2) cognitive difficulties, (3) general health problems, (4) motor symptoms, and (5) functional status.

Methods: A cross-sectional survey study was conducted using the ResearchMatch platform. The key independent variable was age of first exposure to football (AFE < 12 versus AFE ≥ 12). Main outcomes included depressive symptoms (Patient Health Questionnaire-9; PHQ-9), anxiety symptoms (Generalized Anxiety Disorders-7; GAD-7), cognitive difficulties (British Columbia Cognitive Complaints Inventory; BC-CCI), Neurobehavioral Symptom Inventory (NSI) score, and prevalence of other health problems. Multivariable regressions were assessed for associations between AFE and outcome variables.

Results: In total, 107 male participants with exposure to football (mean age: 60.6 ± 15.1 years) reported an average of 4.2 ± 2.7 years of exposure to football, with an average AFE of 11.7 ± 3.1 years. In multivariable analyses, AFE < 12 was not a significant predictor of PHQ-9 (unstandardized beta, B: 0.51, standard error, SE: 1.25, $p = 0.682$), GAD-7 (B: 0.09, SE: 0.95, $p = 0.926$), NSI (B: - 0.56, SE: 2.93, $p = 0.850$), or BC-CCI (B: - 0.65, SE: 0.77, $p = 0.403$). However, more prior concussions were associated with worse PHQ-9 (B: 0.44, SE: 0.10, $p < 0.001$), GAD-7 (B: 0.33, SE: 0.07, $p < 0.001$), NSI (B: 1.04, SE: 0.23, $p < 0.001$), and BC-CCI scores (B: 0.26, SE: 0.06, $p < 0.001$). AFE < 12 did not predict general health problems or independent functional status.

Conclusions: AFE to football was not associated with adverse psychiatric, cognitive, neurobehavioral, or general health outcomes among young, former amateur American Football players. However, more lifetime concussions were associated with adverse cognitive and psychiatric health outcomes. Future studies should examine similar outcomes in older cohorts with more comorbidities to further minimize potential confounding between general health and lack of later-life symptoms.

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The impact of localized muscle mass on sprint mechanics during the swing and stance phases of Division 1 collegiate American football players. Journal of Strength and Conditioning Research, Ahead of Print: 1-5, 2025.

Abstract - This study evaluates the impact of localized muscle volume on sprint mechanics among Division-I collegiate American football players across 3 position groups: Skills (Defensive Backs, Wide Receivers), Mids (Linebackers, Running Backs, Tight Ends), and Bigs (Offensive, Defensive Linemen). A total of 108 male athletes (age: 21.4 ± 1.9 years, BMI: 29.6 ± 4.3) were assessed. Lower extremity muscle volumes were measured using magnetic resonance imaging, and sprint kinematics were captured using an inertial motor unit system during maximal effort sprints. Stepwise multiple linear regressions were conducted to evaluate the relationship between muscle volumes, stance, and swing times for each position group. In the Skills group, increased volumes in the vastus intermedius ($\beta = -0.15$, $p < 0.01$) and sartorius ($\beta = -0.17$, $p = 0.01$) were associated with reduced swing time, while those in the vastus medialis ($\beta = 0.18$, $p < 0.01$) increased swing time. For stance time, the gluteus maximus ($\beta = -0.02$, $p = 0.03$) was associated with reduced stance time, and the tibialis posterior ($\beta = 0.28$, $p < 0.01$) increased it. In the Mids group, the biceps femoris long head ($\beta = -0.15$, $p < 0.01$) was associated with reduced stance time, while the vastus medialis ($\beta = 0.09$, $p < 0.01$) increased it. In the Bigs group, the rectus femoris and sartorius were associated with reduced stance time, while the tensor fasciae latae and tibialis anterior increased stance time. These findings highlight the role of localized muscle volume in influencing sprint mechanics and emphasize the potential need for position-specific physical development programs tailored to the biomechanical demands of running mechanics in American football players.

Practical Applications - This study emphasizes the need for targeted strength and conditioning programs to optimize sprint mechanics in collegiate American football players. For Skill players, exercises that target the sartorius and vastus intermedius, such as lunges and hip flexor strengthening, can help reduce swing time and improve leg recovery. Conversely, careful programming is necessary to prevent excessive development of the vastus medialis, which may contribute to increased swing time. Although front squats preferentially target the quadriceps including the vastus medialis, their volume and intensity should be managed appropriately within a balanced training program to avoid excessive hypertrophy in this region. For Mids and Bigs, focusing on exercises that enhance the biceps femoris long head and rectus femoris, such as Nordic hamstring curls and Romanian deadlifts, may aid in reducing stance and swing times. In addition, incorporating plyometrics improves rate of force

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development, ensuring functional power output (4). Furthermore, this may aid in the transfer of training effect leading to improved on-field performance (2). Finally, incorporating this information into tailored, position-specific training strategies is essential to optimize sprint performance across different positional demands, ensuring the potential for high performance.

The influence of limb dominance on performance-based outcomes after ACL reconstruction. Sport Health, Ahead of Print: 1-8, 2025.

Background: Considering limb dominance (LD) may be valuable when utilizing limb symmetry index (LSI) when assessing patients after anterior cruciate ligament reconstruction (ACLR).

Hypothesis: Patients will have better performance-based outcomes when index ACLR occurred on the dominant limb (DL) compared with the nondominant limb (NDL).

Study design: Observational cross-sectional study.

Level of evidence: Level 3.

Methods: A total of 279 patients (49.1% female, 20.83 ± 5.23 years; 48% DL surgery) completed a laboratory visit (7.65 ± 1.65 months post-ACLR) assessing LD influence on knee extension and flexion peak torque, center of pressure (COP) distance and velocity, and hop performance. LD was defined as the preferred limb to kick a soccer ball. LSI variables were converted to indicator variables and categorized as "Pass" or "Fail" based on LSI ≥ 90%.

Results: Patient LD and isometric knee extension LSI Pass status were significantly associated (χ^2 = 10.09; P = 0.001). No additional associations were found between LD and other LSI Pass status variables (P > 0.05). Patients with DL ACLR demonstrated more symmetric knee extension peak torque (P < 0.001, d = 0.42) and 6-meter hop (P = 0.02, d = 0.25) outcomes; NDL was more symmetric during COP distance tests (P = 0.03, d = 0.40). No differences were observed between LD and raw strength or balance measures (P > 0.05). Patients with ACLR on their NDL jumped farther on their contralateral limb for triple-hop (P = 0.03, d = 0.23) but not single-hop (P > 0.05) distance.

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Conclusion: LD appears to influence isometric knee LSI in patients post-ACLR; patients with surgery on their DL achieved higher symmetry and a greater rate of LSI Pass success for knee extension strength.

Clinical relevance: Addressing differences in recovery patterns between DL and NDL may improve rehabilitation precision and guide return-to-activity timelines after ACLR.

The relationship between relative rest and team performance across competitive seasons in the National Football League. Journal of Strength and Conditioning Research, Ahead of Print: 1-5, 2025.

Abstract - This investigation aimed to assess the relationship between relative rest index (RRI) and team performance across multiple seasons of National Football League (NFL) competition. To examine this relationship, all NFL regular season games from the 1970-71 to 2022-23 NFL seasons (N = 12,290) were analyzed with rest intervals calculated for each team on a game-to-game basis as the number of days between current and previous games. Next, RRI was calculated as the discrepancy in rest intervals between competing teams for all games and averaged for all teams across all seasons with teams categorized by season average RRI (<-0.70; -0.7 to -0.5; -0.5 to -0.25; -0.25 to 0; 0 to 0.25; 0.25 to 0.5; 0.5 to 0.7; >0.70). Win percentage and playoff depth (lost in wild card round; lost in divisional round; lost in conference championship; lost in Super Bowl; won Super Bowl; did not make playoffs) quantified team performance. Pearson product correlations were implemented to assess the relationship between season average RRI and win percentage. One-way analyses of variance assessed differences in win percentage across RRI categories and RRI across playoff depth categories. A trivial negative correlation between RRI and win percentage across all seasons was noted. Moreover, win percentage did not differ across RRI categories, while RRI was attenuated among teams that lost in the Super Bowl and divisional round compared with teams that did not qualify for the playoffs, suggesting that RRI has no significant effect on team performance with competitive eras of NFL competition collated.

Practical Applications - These data demonstrate that RRI does not significantly affect Win/Loss Ratio across all seasons of NFL competition from 1970 to the present day. In fact, teams progressing further in competition displayed reduced RRI. As such, the present scheduling configuration used by the NFL seems to maintain competitive equity. However, this analysis focused solely on team performance.

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Coaches and practitioners should not disregard or underprioritize player recovery from both practice and competition because of the robust demands imposed by an American Football season.

The role of high-intensity interval training (HIIT) in neuromuscular adaptations: Implications for strength and power development – A review. Life, 15: 657, 2025.

Abstract - High-intensity Interval Training (HIIT) is increasingly recognized for enhancing neuromuscular function, strength, power, and overall athletic performance. This review systematically examined peer-reviewed studies published between 2000 and 2025, focusing on HIIT's impact on motor unit recruitment, muscle fiber composition, neuromuscular efficiency, maximal strength, rate of force development (RFD), muscle hypertrophy, and power output. Findings indicate that HIIT significantly improves neuromuscular activation by increasing motor unit recruitment and synchronization, particularly in fast-twitch fibers essential for explosive movements. HIIT also promotes shifts toward Type II and hybrid Type IIa fibers, enhancing strength and endurance. However, despite its effectiveness in boosting RFD and explosive power, HIIT is less efficient than traditional resistance training for maximizing absolute strength and hypertrophy due to insufficient progressive overload. Integrating resistance-based and plyometric-based HIIT protocols emerged as an effective strategy to enhance strength and power concurrently. Effective periodization and recovery strategies, including active recovery and targeted nutrition, help mitigate fatigue and optimize adaptations. Notable research gaps include the long-term impacts of HIIT on neuromuscular function and the efficacy of individualized HIIT protocols based on specific athlete characteristics. Future research should focus on refining HIIT protocols for different sports, exploring its synergy with traditional resistance training, and assessing long-term adaptations to sustain strength and power gains. HIIT presents a valuable, time-efficient complement to conventional training methods for improving strength, power, and neuromuscular efficiency.

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Training impulse and session ratings of perceived exertion as perceived by athletes and observed by coaches during preseason football camp. Journal of Strength and Conditioning Research, Ahead of Print: 1-9, 2025.

Abstract - Quantification of internal training load (TL), through wearable technology, represents an emerging metric for TL management in sport. The accuracy of subjective vs. objective assessments of TL, as perceived by athletes and observed by coaches, has yet to be evaluated during long-duration football practices (>2 hours). The purpose of this study was to evaluate relationships among objective heart rate (HR) and subjective ratings of perceived exertion (RPE) assessment of TL in collegiate American football athletes, and their coaches, during preseason camp. Twenty-four athletes (first-year, n = 6; returning, n = 18) and 2 coaches (football coach [FBC], n = 1; strength and conditioning coach [SCC], n = 1) participated. Athletes wore chest strap HR monitors during the first 6 days of preseason football camp for an objective TL assessment using training impulse (TRIMP: average exercise HR × duration of practice). Subjective TL was assessed by both athletes (perceived) and coaches (observed) using a session ratings of perceived exertion (sRPE; 0-10 rating scale; 10 = maximum effort). The average practice duration was 142.3 ± 35.5 minutes, with 139 total observations across 6 practices. Significant (alpha level $p \leq 0.05$) relationships were noted between TRIMP and sRPE from the athletes ($r = 0.62$), FBC ($r = 0.68$) and SCC ($r = 0.49$). When subdivided into first-year and returning groups, relationships were stronger between TRIMP and sRPE with first-year athletes (athlete $r = 0.68$; $p = 0.001$; FBC $r = 0.79$; $p = 0.001$; SCC $r = 0.64$; $p = 0.001$) than for returning athletes (athlete $r = 0.60$; $p = 0.001$; FB coach $r = 0.70$; $p = 0.001$; SCC $r = 0.40$: $p = 0.001$). These data suggest that subjective athlete perceptions and coach observations of TL are in low-moderate agreement, indicating these are not interchangeable, during long-duration preseason football camp.

Practical Applications - The overarching purpose of this study was to determine whether heart-rate monitoring systems enhanced our ability to assess individual athlete effort and recovery; particularly after long grueling football practices conducted during the peak summer heat. It seems that both objective (HR) and subjective (sRPE) measures of effort and recovery likely represent independent—but interdependent—modulators of stress. With practice and feedback about athletes' heart rates, both athletes and coaches may improve their TL perceptions and observations, respectively; such that their TL reports would align more closely with objectively measured TL. Athletes and coaches should continue

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to assess both objective and subjective measures of internal load, as both likely contribute unique insights into multiple stressors, which affect athlete load plus subsequent management.

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